



Notes & Tips

IQPA: Isothermal nucleic acid amplification-based immunoassay using DNzyme as the reporter system



Qiuting Loh^a, Noorsharmimi Omar^a, Jörn Glöckler^b, Theam Soon Lim^{a,*}

^a Institute for Research in Molecular Medicine, Universiti Sains Malaysia, Minden, Penang 11800, Malaysia

^b Department of Molecular Biotechnology and Functional Genomics, Technical University of Applied Sciences Wildau, Wildau 15745, Germany

ARTICLE INFO

Article history:

Received 4 April 2014

Received in revised form 12 June 2014

Accepted 12 June 2014

Available online 24 June 2014

Keywords:

Biosensors

DNzyme

Antibody–antigen detection

Isothermal amplification

ABSTRACT

Immunoassays are often coupled to peroxidase activity for antigen detection. Sensitivity and speed of detection has been increased by the advent of hybrid methods such as immuno-PCR (polymerase chain reaction). However, a more simplified immunoassay that retains both colorimetric peroxidase detection and effective DNA amplification in a setting closer to field application conditions has been nonexistent. Here we describe a method that successfully combines a competitive immunoassay with the new isothermal quadruplex-primed amplification (QPA) to generate excess quadruplex reporter molecules with intrinsic peroxidase DNzyme activity.

© 2014 Elsevier Inc. All rights reserved.

In conjunction with the initiative proposed guidelines, namely ASSURED (affordable, sensitive, specific, user-friendly, robust and rapid, equipment free, and deliverable),¹ we have developed an alternative immunoassay method, namely IQPA, in which the detection is based on DNA molecules that are formed after isothermal amplification. For our assay, we chose a sandwich format to allow a fast but sensitive assay that is easily applicable for the detection of molecules relevant for field-testing such as antibiotics and toxins in a competitive manner. During recent years, new detection methods other than antibody–enzyme conjugates have been developed to improve sensitivity or specificity. Among these, antibody–DNA conjugates have proven to be very sensitive in a so-called immuno-PCR (polymerase chain reaction) format [1–3]. In general, antibodies are directly or indirectly conjugated to oligonucleotides that serve as templates for subsequent PCR amplification and analysis. Furthermore, the combination with real-time PCR allows both sensitive and quantitative detection of the target analytes. However, both afford the use of thermocyclers that, due to their size and energy demands, are not compatible with the field experiments.

In comparison with PCR and other amplification methods, isothermal quadruplex-primed amplification (QPA) has the advantage

of employing only a minimal set of primers and enzymes while retaining stringent conditions for specific target amplification [4,5]. QPA can be used in both linear and exponential amplification formats, depending on the amount of primer binding sites present in the given template. The singular temperature requirement without any cycling allows the use of a simple incubator that can be designed in a miniaturized and portable format.

The detection of QPA product formation has been previously achieved by incorporation of labeled nucleotides that fluoresce differentially, depending on their conformation, as either disordered single-strand primer or G-quadruplex amplification product [6]. In principle, many additional detection methods are possible, most also involving the use of expensive labels.

Because QPA already results in the accumulation of G-quadruplexes and many oligonucleotides forming such structures are known to exhibit peroxidase-mimicking activity [7–9], we were interested in whether a combination of QPA with an immunoassay was possible. Ideally, the detection is based on the colorimetric peroxidase assay that can be qualitatively evaluated by the naked eye. This article describes the fusion of QPA to an immunoassay platform for the detection of antibody–antigen interaction. The method takes advantage of the isothermal nature of QPA to amplify G-quadruplex-forming DNA to be used as a direct reporter system for the immunoassay. The method, coined as immuno-QPA (IQPA), provides an alternative to the conventional immunoassay methods and eliminates the use of chemically conjugated enzymes as reporter systems.

Thus, we first analyzed the DNzyme activity of the QPA quadruplex in the colorimetric peroxidase assay based on ABTS

* Corresponding author. Fax: +60 4 653 4803.

E-mail address: theamsoon@usm.my (T.S. Lim).

¹ Abbreviations used: ASSURED, affordable, sensitive, specific, user-friendly, robust and rapid, equipment free, and deliverable; PCR, polymerase chain reaction; QPA, quadruplex-primed amplification; IQPA, immuno-QPA; ABTS, 2,2'-azino-bis-3-ethylbenzothiazoline-6-sulfonic acid; RT, room temperature; HRP, horseradish peroxidase; ELISA, enzyme-linked immunosorbent assay; RCA, rolling circle amplification; qPCR, quantitative PCR.

(2,2'-azino-bis-3-ethylbenzothiazoline-6-sulfonic acid). Next, we further optimized the assay to buffers and conditions required for efficient isothermal QPA (see online [Supplementary Material](#)). Then, we combined the antibody sandwich assay with an oligonucleotide for isothermal amplification, resulting in peroxidase activity of the amplified DNAzyme that is detected by ABTS turnover ([Supplementary Fig. 1](#)).

The detailed protocol of the sandwich assay and the oligonucleotide design for QPA DNA template and primer are shown in the [Supplementary Material](#). After the unbound streptavidin was washed away, 1 μ M of the biotinylated QPA exponential DNA template was introduced to the well for 1 h, followed by the washing of unbound DNA template. In the same well, quadruplex priming amplification reactions were conducted in a 50- μ l reaction mixture containing 20 mM Tris-HCl (pH 8.8), 10 mM $(\text{NH}_4)_2\text{SO}_4$, 0.1 mg/ml bovine serum albumin (BSA), 0.1% (v/v) Triton X-100, 2 mM MgSO_4 , 0.5 mM deoxynucleoside triphosphate (dNTP), and 1.25 U of *Vent* DNA polymerase (New England Biolabs, Ipswich, MA, USA) and 10 μ M primer. The mixture was heated at 95 $^\circ\text{C}$ for 2 min using a hot plate, and then KCl was added into the mixture to a final concentration of 25 mM and the well plate was incubated in the oven at 75 $^\circ\text{C}$ for 1 h. Following amplification, 5 μ l of the 5 mM hemin was added into the mixture and incubated for 1 h at room temperature (RT) in the dark. Optionally, the mixture was transferred to a tube with 5 mg of activated charcoal (common charcoal used for a fish aquarium) and left at RT for 10 min, followed by centrifugation at the highest speed for 2 min to remove the charcoal. Then, the supernatant was transferred to another tube containing 100 μ l of ABTS solution with 10 μ l of 100% H_2O_2 . After 1 h of incubation with hemin, the absorbance reading of the sample was taken using a Thermo Scientific Multiskan spectrophotometer (Waltham, MA, USA).

The direct comparison of the known hemin aptamer [10] with the QPA quadruplex resulted in approximately 30% activity of the latter, which was deemed to be sufficient for the assay ([Fig. 1A](#)). However, we noticed that there was a concomitant reaction as a result of the interaction between hemin and ABTS without any quadruplex template. This is mainly due to hemin itself having the ability to catalyze ABTS in the presence of H_2O_2 . Therefore, without the sensitive instruments to measure the color intensity, it is doubtful that the QPA-DNAzyme reaction can be detected by the naked eye. Thus, absorption measurements were performed to quantify the significant color intensity difference between samples with DNAzyme formed and also sample without DNAzyme formed or with hemin alone.

To apply a visual readout with low background, we investigated the use of activated charcoal to improve the detection. We anticipated that the activated charcoal would absorb the unbound hemin, leaving only bound hemin within the quadruplex to be reacted with ABTS. Overall, we noticed that the sample readout was also reduced as a result of adding in activated charcoal. This shows that the activated charcoal not only absorbs free hemin but also is capable of absorbing quadruplex-bound hemin. The overall reduction in signal was universal for both free hemin and bound hemin samples, with a lower reduction for quadruplex-bound hemin. The higher reduction in signal for free hemin could be due to a faster rate of absorption for free hemin by activated charcoal compared with the quadruplex-bound hemin. Therefore, prolonged incubation with charcoal can result in a total reduction of the color intensity from the ABTS reaction.

By adding activated charcoal to the reaction, the difference in ABTS turnover could be increased for visualization by the naked eye ([Fig. 1B](#)). We found that application of the charcoal does not increase the sensitivity of the assay given that the difference between the sample readout with and without charcoal is similar after deducting the background intensities. Thus, the use of

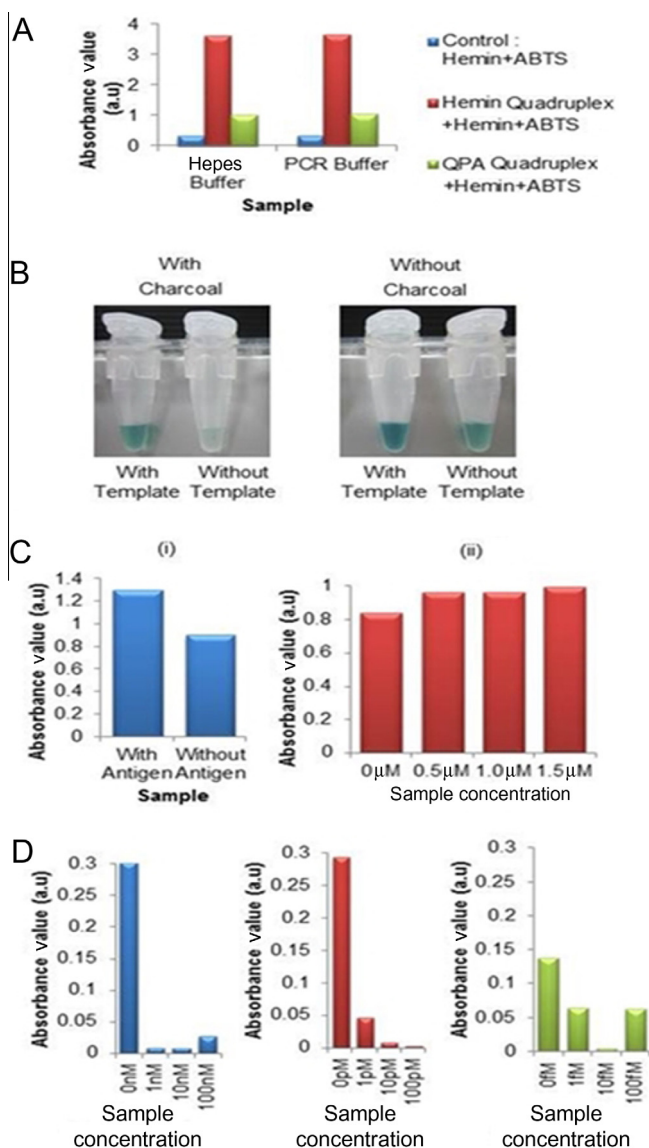


Fig. 1. Colorimetric change of ABTS due to the formation of DNAzyme by QPA. (A) Absorbance value of hemin quadruplex and QPA quadruplex forming DNAzyme in Hepes buffer and PCR buffer. (B) Comparison of absorbance values of the sample with and without the presence of activated charcoal. Photographs of oxidation of ABTS by H_2O_2 in the presence of DNAzyme produced by QPA with and without activated charcoal are shown. (C) (i) Absorbance value of samples with and without antigen added in the direct antibody-antigen detection assay. (ii) Titration curve of the concentration of biotinylated antigen for direct immunoassay. (D) Titration curves of the concentration of non-biotinylated antigen for competitive assay based on IQPA (i) and conventional ELISA (ii).

activated charcoal in this assay does not offer major advantages if proper instrumentation is available. The use of activated charcoal is recommended only if evaluation by the naked eye is intended.

As shown in [Fig. 1C\(i\)](#), the absorbance value for the sample with the antigen bound is comparatively higher than the sample without the antigen. Because the biotinylated antigen was bound to the antibody coated in the well, this allowed the streptavidin to be sandwiched in between the antigen and the biotinylated DNA template. During QPA, quadruplexes were formed and afterward bound with hemin to produce DNAzymes. The DNAzymes were reacted in the ABTS solution, resulting in the colorimetric change of the ABTS solution. When the antigen was absent in the assay, an elevated background reading was detected. This might be due to the free hemin that readily reacts with H_2O_2 in the ABTS

solution. Nevertheless, this is also common with most enzymatic assays such as horseradish peroxidase (HRP)-based colorimetric assays in immunoassays.

In competitive immunoassay, the absorbance value decreases as the concentration of the sample antigen (in this case, the non-biotinylated antigen) increases. The limit of detection for the IQPA-based competitive assay was obtained. As shown in Fig. 1D(i), the result indicated that this competitive IQPA can detect as low as 1 fM concentration of the sample antigen. Therefore, the sensitivity of the competitive IQPA is comparable to that of the conventional competitive enzyme-linked immunosorbent assay (ELISA) even though the absorbance values are not as high as the absorbance values of the conventional ELISA (Fig. 1D(ii)). Nevertheless, competitive IQPA will be useful if the secondary enzyme-conjugated antibody needed in the conventional ELISA is not available. In addition, a titration curve for the concentration of the biotinylated antigen in the IQPA-based direct assay was also plotted (Fig. 1C(ii)). However, the direct IQPA can detect only as low as 0.5 μ M of the antigen, which is much less sensitive than the competitive assay.

Many forms of nucleic acid amplification methods have been incorporated successfully into various immunoassays [1,2]. The amplified products are basically the reporter system of the assay, but most require the physical conjugation of reporter dyes or DNA chelating dyes. In this study, we successfully developed an alternative immunoassay method by incorporating QPA into the assay. It involves isothermal amplification of a quadruplex-forming DNA template that is a single-stranded DNA. There have been many DNAzyme-based immunoassays reported; however, most use a DNAzyme-linked aptamer that is specially designed to be bound to the target protein. In this assay, for the DNA template to be bound to the protein, we employed the simple streptavidin–biotin interaction whereby streptavidin was added into the assay sandwiched between the target biotinylated antigen and the biotinylated DNA template. The DNA amplification process in IQPA was carried out at a single temperature, with immuno-rolling circle amplification (RCA) being the only other method reported. Immuno-RCA was carried out at 37 °C, whereas IQPA was carried out at 75 °C. The higher temperature is a favorable feature because it can help to remove nonspecific amplification during the assay.

The experimental design of IQPA is suitable to meet the criteria of the ASSURED initiative. The requirement of sophisticated equipment can be avoided for IQPA because it requires only an oven for amplification. The typical immuno-PCR requires expensive instrumentation for thermal cycling and detection such as a thermocycler or a quantitative PCR (qPCR) machine. Moreover, IQPA is a colorimetric-based immunoassay that can be carried out with detection by the naked eye. It can generate DNAzyme that catalyzes the oxidation of ABTS, whereas other enzymatic immunoassays based on colorimetric changes require HRP to catalyze the reaction. For non-enzymatic immunoassays such as the typical immuno-PCR or immuno-RCA, the samples obtained from immuno-PCR will be analyzed by running them on gel electrophoresis, which is time-consuming and cumbersome. Some applications require real-time fluorescence detection, whereby a qPCR machine is needed for sample analysis. Many other studies require

fluorescence dye that is probed on the DNA, which again requires fluorescence readers for detection. Therefore, IQPA is a useful alternative for simple visualization-based assay that can be applied effectively in a low-resource setting.

Lastly, IQPA is best considered as a qualitative assay and not a quantitative assay. The end product of IQPA is based on the amplification of the quadruplex-forming DNAzyme; therefore, the efficiency of the amplification step and its product could not be rigorously controlled and maintained consistent in independent samples. It remains to be seen whether other quadruplex sequences can be found that provide a better combination for both efficient amplification and DNAzyme activity in assay conditions. An important source of variability that is intrinsic to isothermal amplification is lack of synchronization of polymerase activity, in contrast to PCR that is based on thermocycling. In its current form, IQPA cannot generate the reproducible result that is required for a quantitative assay. However, taking together all of its advantages, this new technique can be an appealing choice for protein–protein interaction or detection assays.

Acknowledgments

The work was supported by a Malaysian Ministry of Education through the Higher Institution Centre of Excellence (HiCoE) grant (311/CIPPM/44001005) and a Universiti Sains Malaysia Short Term Grant (304/CIPPM/6312060).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ab.2014.06.012>.

References

- [1] C.M. Niemeyer, M. Adler, R. Wacker, Immuno-PCR: high sensitivity detection of proteins by nucleic acid amplification, *Trends Biotechnol.* 23 (2005) 208–216.
- [2] C.M. Niemeyer, M. Adler, B. Pignataro, S. Lenhart, S. Gao, L. Chi, H. Fuchs, D. Blohm, Self-assembly of DNA–streptavidin nanostructures and their use as reagents in immuno-PCR, *Nucleic Acids Res.* 27 (1999) 4553–4561.
- [3] H. Zhou, R.J. Fisher, T.S. Papas, Universal immuno-PCR for ultra-sensitive target protein detection, *Nucleic Acids Res.* 21 (1993) 6038–6039.
- [4] B.I. Kankia, Self-dissociative primers for nucleic acid amplification and detection based on DNA quadruplexes with intrinsic fluorescence, *Anal. Biochem.* 409 (2011) 59–65.
- [5] A. Taylor, A. Joseph, R. Okyere, S. Gogichaishvili, K. Musier-Forsyth, B. Kankia, Isothermal quadruplex priming amplification for DNA-based diagnostics, *Biophys. Chem.* 171 (2013) 1–8.
- [6] J. Johnson, R. Okyere, A. Joseph, K. Musier-Forsyth, B. Kankia, Quadruplex formation as a molecular switch to turn on intrinsically fluorescent nucleotide analogs, *Nucleic Acids Res.* 41 (2013) 220–228.
- [7] X.H. Zhao, L. Gong, X.B. Zhang, B. Yang, T. Fu, R. Hu, W. Tan, R. Yu, Versatile DNAzyme-based amplified biosensing platforms for nucleic acid, protein, and enzyme activity detection, *Anal. Chem.* 85 (2013) 3614–3620.
- [8] Z. Zhou, Y. Du, L. Zhang, S. Dong, A label-free, G-quadruplex DNAzyme-based fluorescent probe for signal-amplified DNA detection and turn-on assay of endonuclease, *Biosens. Bioelectron.* 34 (2012) 100–105.
- [9] N. Omar, Q. Loh, G. Tye, Y. Choong, R. Noordin, J. Glöckler, T. Lim, Development of an antigen–DNAzyme based probe for a direct antibody–antigen assay using the intrinsic DNAzyme activity of a daunomycin aptamer, *Sensors* 14 (2013) 346–355.
- [10] T. Li, L. Shi, E. Wang, S. Dong, Multifunctional G-quadruplex aptamers and their application to protein detection, *Chem. Eur. J.* 15 (2009) 1036–1042.